

September
10-09-2025
Wednesday

Day - 10

Lecture - 08

CSS Box Model | Inline & Block Elements

* CSS Box Model

```
<h1> Hello Coder </h1>
```

```
h1 {
```

```
background-color: orange;
```

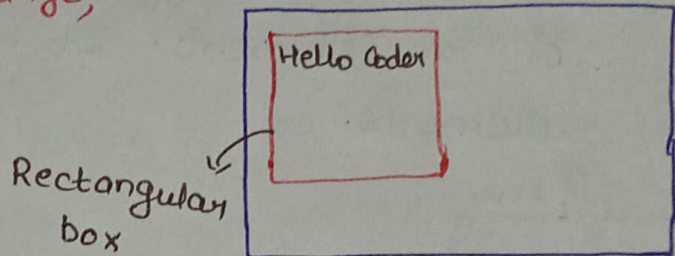
```
height: 300px;
```

```
width: 300px;
```

```
}
```

Can we change the
height and width of <h1>
tag

- Yes



Browser

- Border: Border is a line around the content. It is like frame of box

```
h1 {
```

```
background-color: orange;
```

```
height: 300px;
```

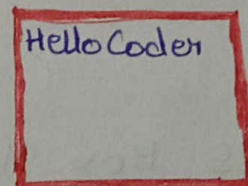
```
width: 300px;
```

```
border-width: 10px;
```

```
border-style: solid;
```

```
border-color: red;
```

```
}
```



we can change the style like
dashed (---) and dotted (....)
and more

→ Or we can write it in any
order

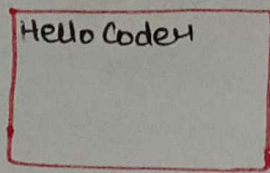
/* Shortcut */

```
border: 10px solid red
```

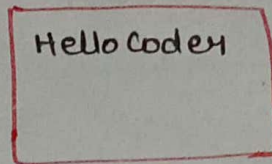
↓ ↓ ↓
width style color

```
border: solid red 10px;
```


- **Padding**: Padding is a space between the content and the border of a box.



Without padding
Content will be
attach to border



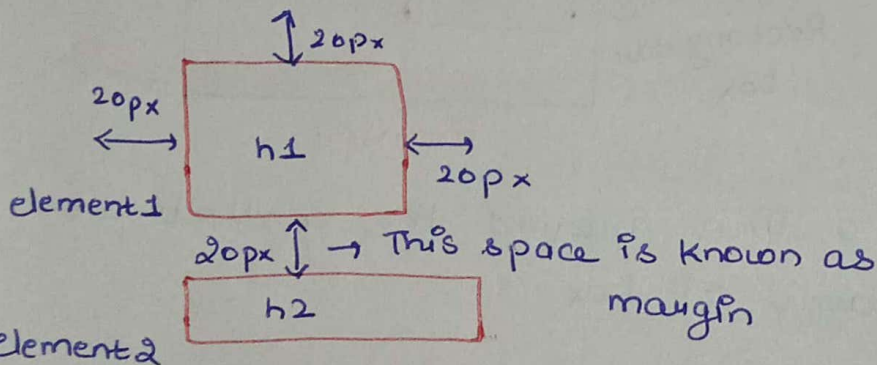
With padding
Content will be
away from border

h2 {

padding: 10px;

} This padding
will be apply
to all the
four sides

- **Margin**: Margin is a space outside the border of an element. It creates space between elements.

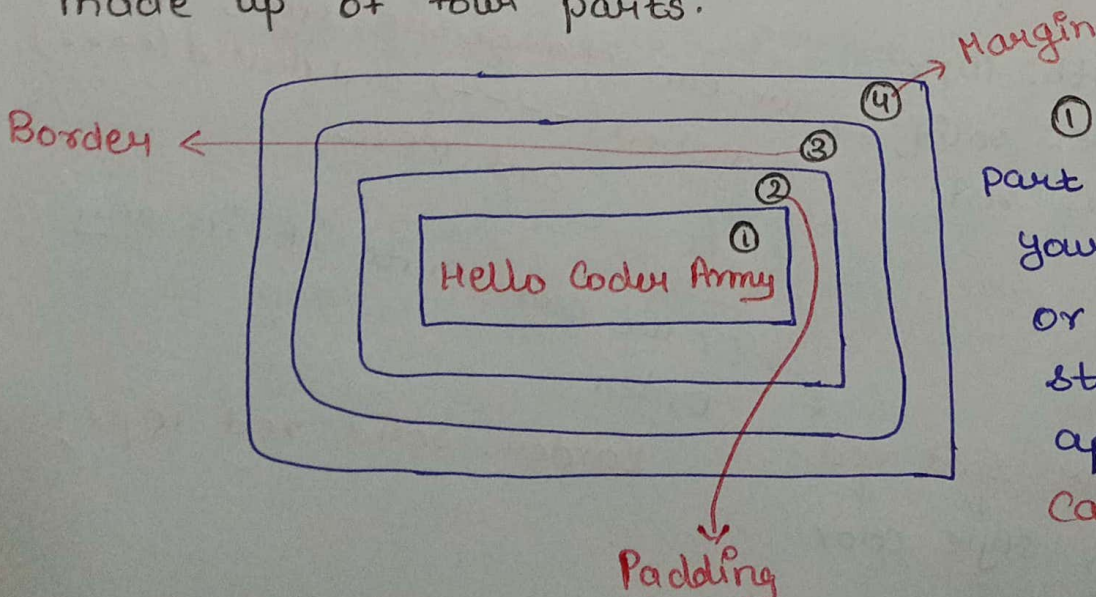


h1 {

margin: 20px;

} This margin will
be apply in all the
direction

So, CSS Box Model explains how every HTML element is treated like a box and the box is made up of four parts.



① Content It is
part of box where
your images, text
or ~~actually~~ other
stuff actually
appear. and we
can fix width
and height

- When we write any tag that will go to rectangular box and that is known as **Content**
- If we add border to content then it will attach with the content so to make distance between them we add **Padding**

You can view the box model by doing inspect on web page

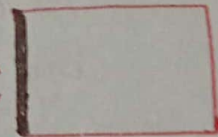
* Individual Properties or Combine (short-hand notation)

We can give individual property to any element by specifying it with name.

• Border :

border-left: 10px solid black;

Apply border only in left direction



border-right: 20px solid black;

• Padding :

We can also give individual padding to any side

padding-left: 10px;

padding-bottom: 20px;

padding: 10px 20px 30px 40px;

↑ ↑ ↑ ↑
top right bottom left

← Shorthand notation to give padding

padding: 10px 20px;

↑ ↑
top & bottom left & right

padding: 10px 20px 30px;
 ↑ ↑ ↑
 top left & right bottom

Similarly you can give margin to individual sides and short-hand notation.

margin-left: 20px;

margin-bottom: 30px;

margin: 10px 20px 30px 40px;
 ↓ ↓ ↓ ↓
 top right bottom left

* Border - radius:

border-radius: 10px;

To smoothness the edge of a border by making it rounded

* Block-level Element vs Inline Element

Inline-Element: Only take up as much width as their content needs.

Example: , <a>,

<p> Lorem ipsum sit amet, </p>

```
span {  
  background-color: pink;  
}
```

→ This pink color only apply to sit word

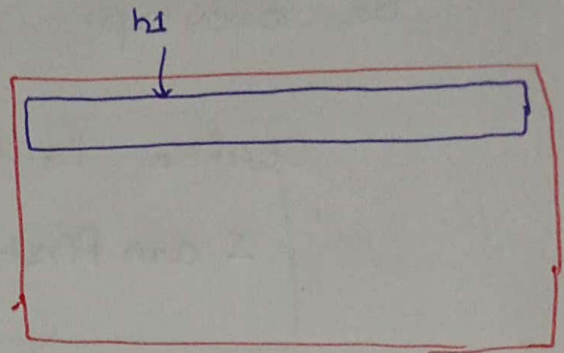
Block - Element: Take the whole width on the page and start with next line always.

`<h1> ... <h1>`

`h1 {`

`background-color: blue;`

`}`



Some random margin padding are available by default by the browser.

`* {`

`margin: 0px;`

`padding: 0px;`

`}`

`*` → Known as universal selector

`h1 {`

`padding: 10px;`

`}`

→ This padding will override the previous 0px

Block - Level

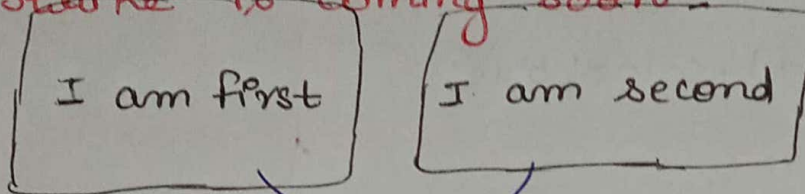
- 1) Take the whole width.
- 2) Always start with new line.
- 3) We can fix height and width
- 4) Margin and padding both are available in all direction

Inline - level

- 1) Take only content width.
- 2) Start with same line and move to next if space is not available.
- 3) We can't apply height and width.
- 4) ^{Imp} Margin and padding will be apply only horizontally not vertically

Note: If you give margin and padding vertically and the background color is applied then it will show you ~~visually~~ ^{visually} and the content will be overlap.

Stake is coming soon



Background color will be visually ~~visually~~ shown but not applicable

How can we make block level to inline element?

```
h1 {  
  display: inline;  
}
```

Now h1 will behave inline and height width not applicable

```
span {  
  display: block;  
}
```

span will behave block level and ~~can~~ height width is applicable

display: inline-block;

Hybrid property

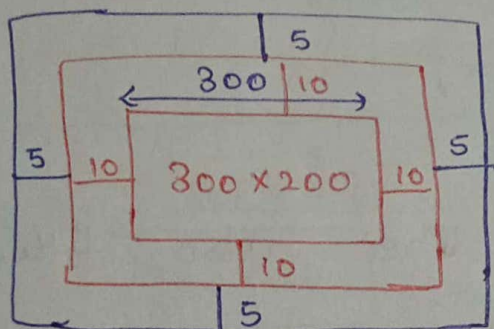
- Behave like inline element
- And we can give height, width, margin - top, bottom
- Means we can set ~~both~~ padding and margin both horizontally and vertically.

* Box - sizing :

- This property is used to define how the total width and height of an element is calculated.
- By default width and height apply to the content - area only

(1) Content - box : default

- the width & height - only to content area
- Padding and border are added outside the content area, increasing total size of the element.



Content - width : 300px

padding : 10px on each side

$$10 + 10 = 20px$$

Border : 5px on each side

$$5 + 5 = 10px$$

$$\begin{aligned} \text{Total - width} &= 300px + 20px + 10px \\ &= 330px \end{aligned}$$

(2) border - box :

- The width and height properties include the content, padding and border

div {

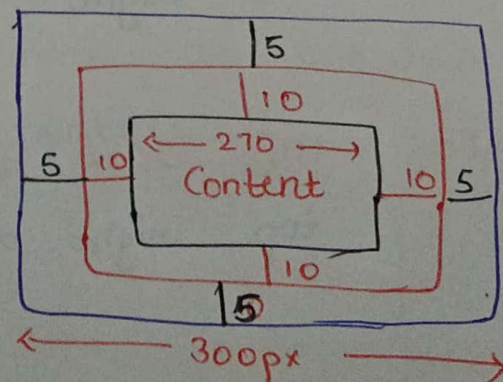
width: 300px;

padding: 10px;

border: 5px solid blue;

box-sizing: border-box;

}



Total width = 300px;

$$\text{padding} = 10\text{px} + 10\text{px} = 20\text{px}$$

$$\text{border} = 5\text{px} + 5\text{px} = 10\text{px}$$

$$\begin{aligned}\text{Content - width} &= 300 - 30 \\ &= 270\text{px}\end{aligned}$$

For Responsive website we use border-box

- Margin will no effect on this
- If we increase padding and border then content-size will be decrease due to box-sizing : border-box;
- If we choose content-box then the content will be spread on the website because it not include padding and border so we have to take care for that